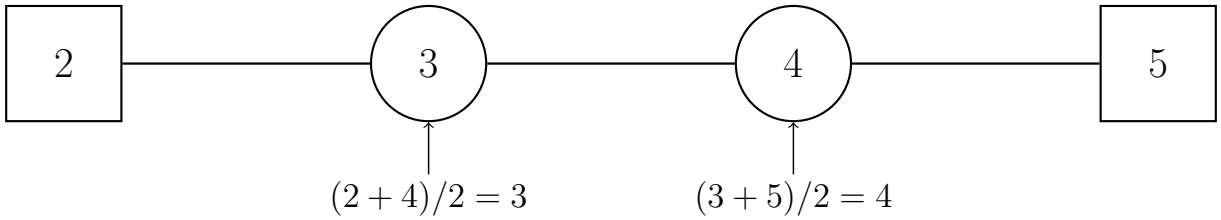
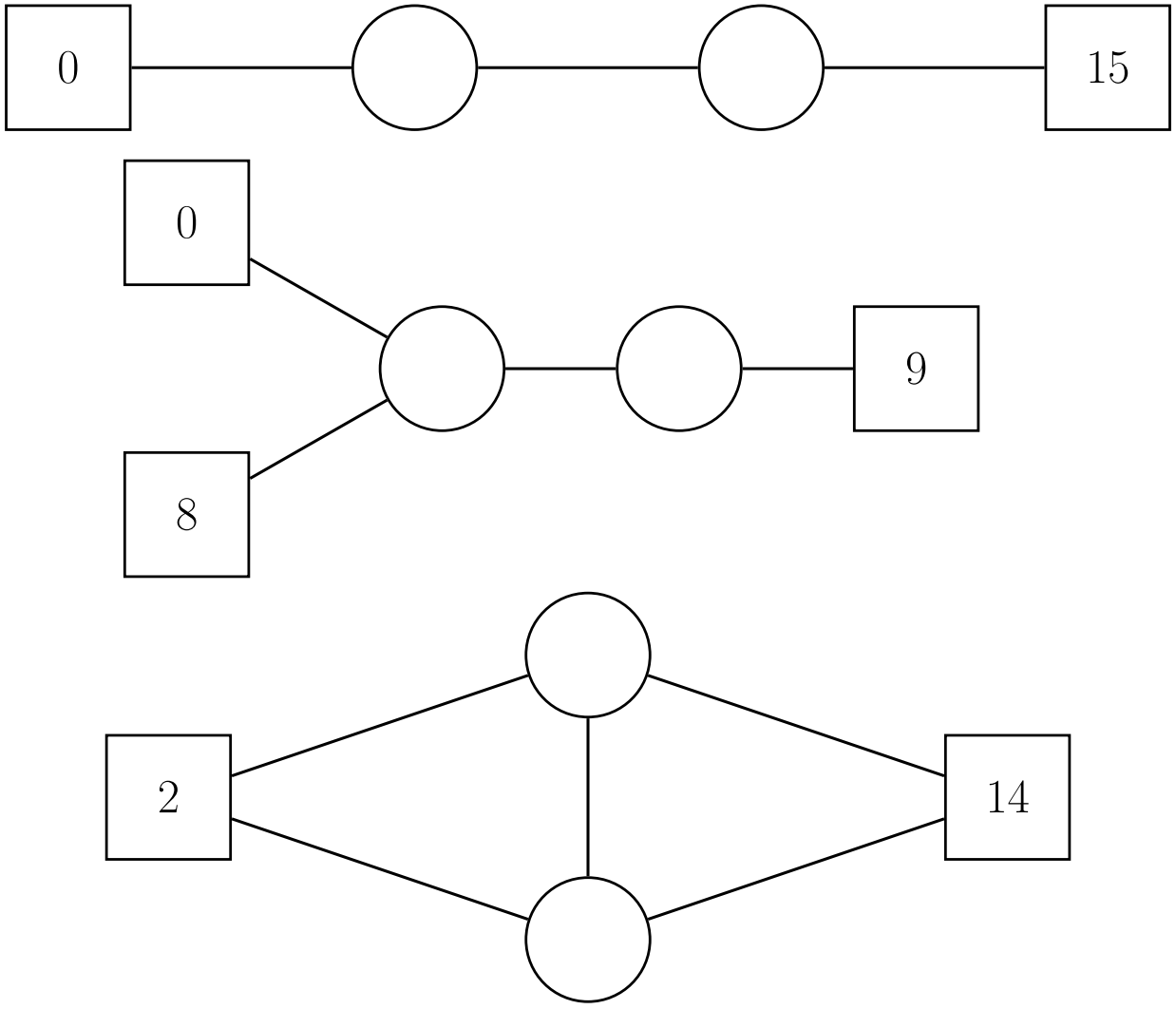
Add the numbers joined to a circle and divide by how many numbers you added. The answer must equal the circle's number. Squares keep their numbers. Every circle must work at the same time.

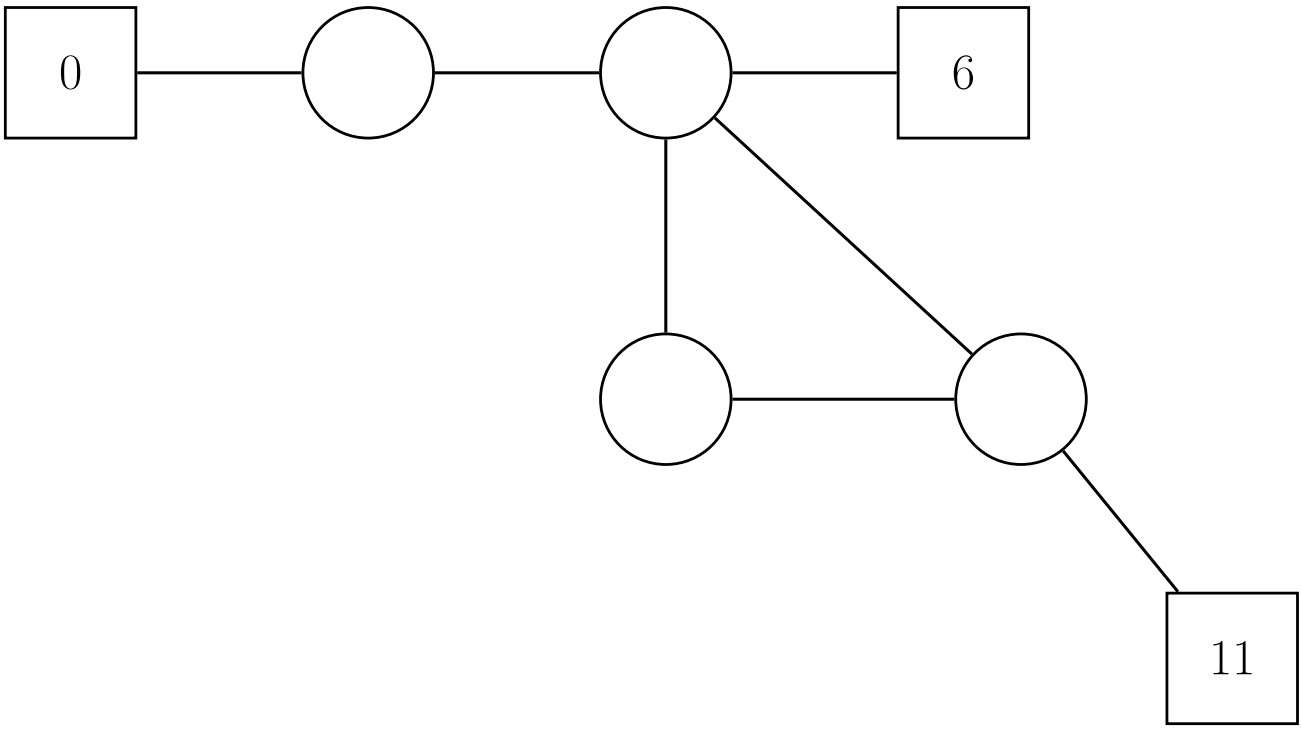
Example: both circles work.

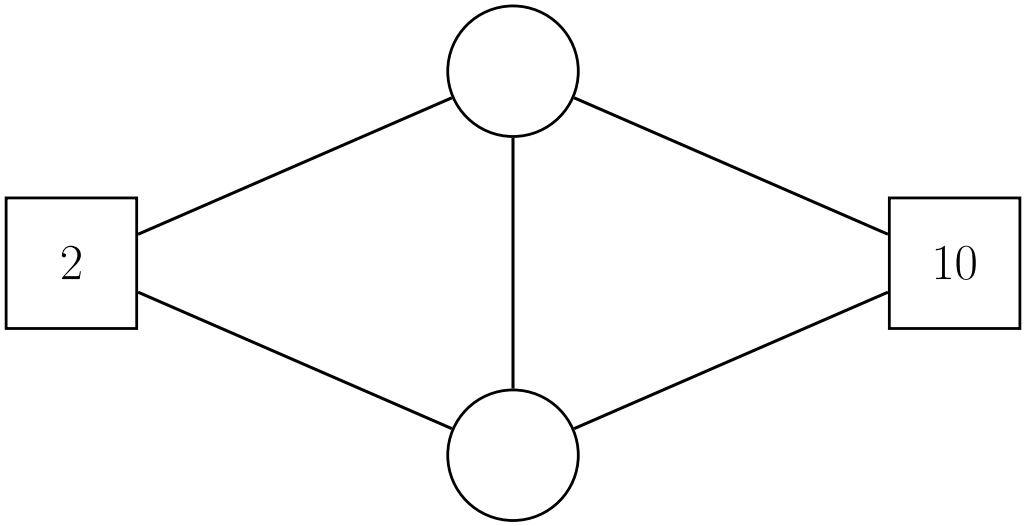


Problem 1: Fill all the circles. Can any board have a second filling with the same square numbers?

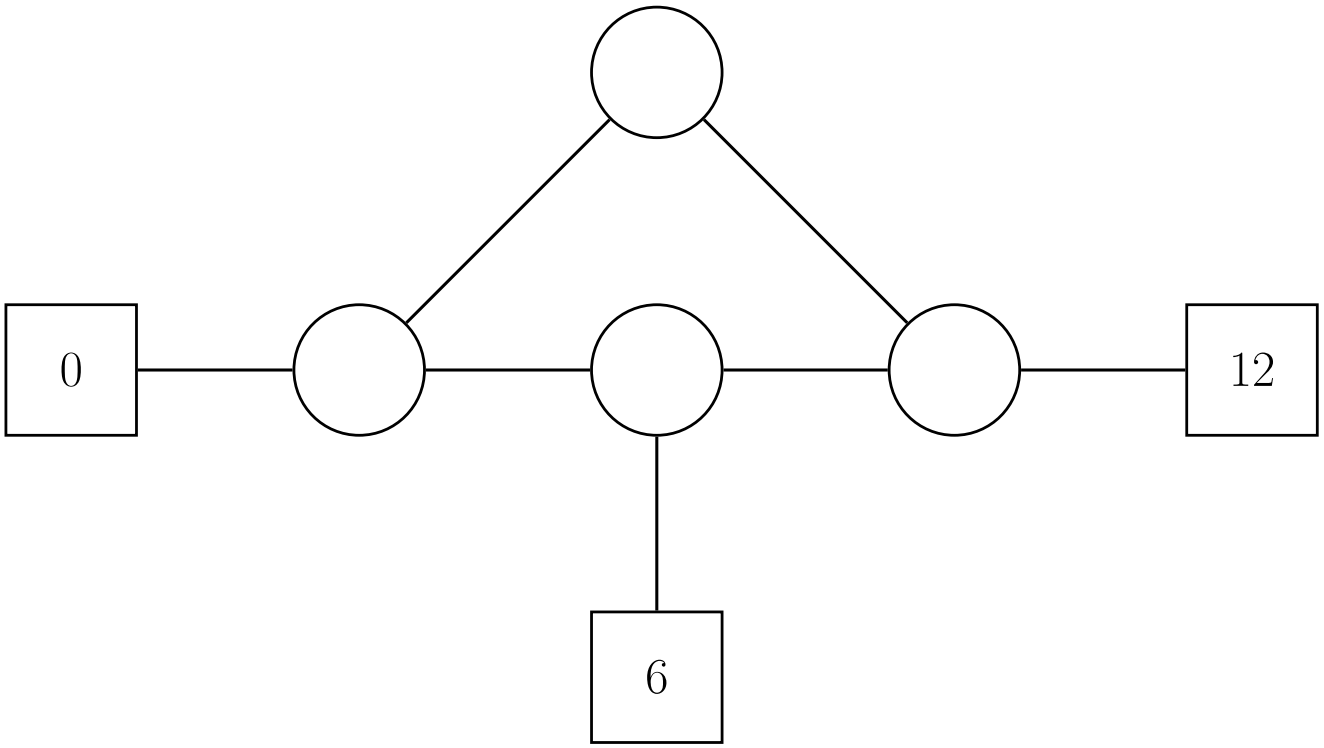


Problem 2: Can either board have a circle number of 13 or more while all circles follow the rule? Find a filling, or explain why none can work.

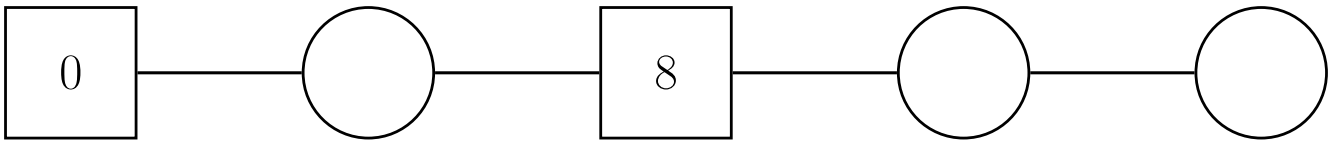


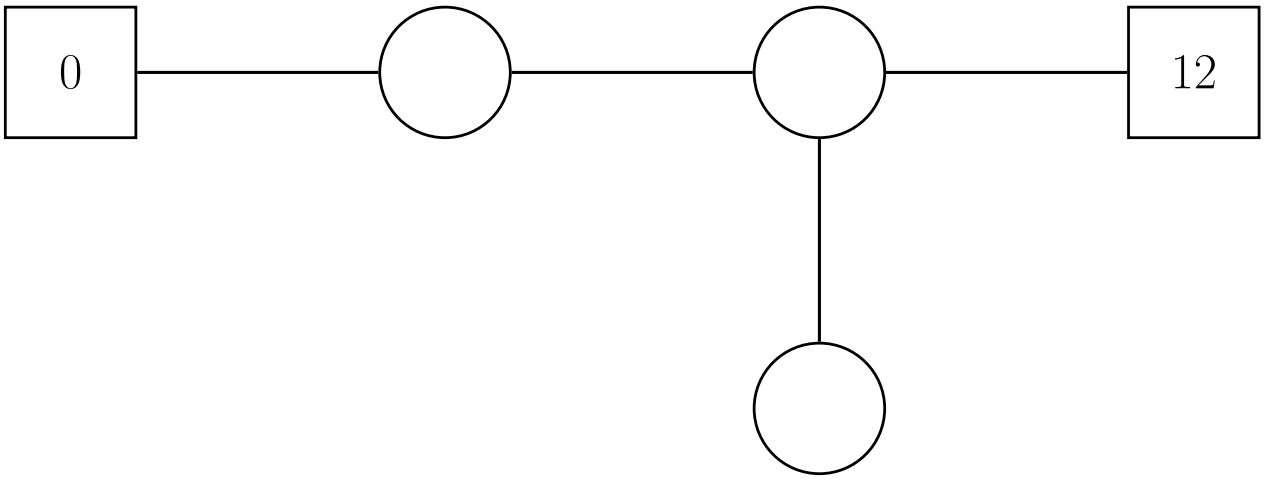


Problem 3: A board has finitely many shapes and lines. Every shape can reach a square along the lines. Explain why no circle number can be below the smallest square number it can reach or above the largest one. Your explanation must work for every such board.

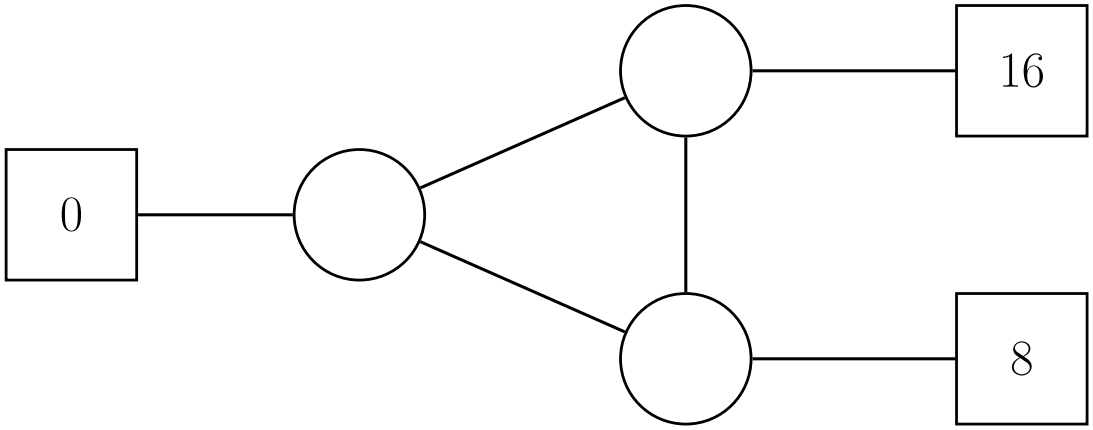


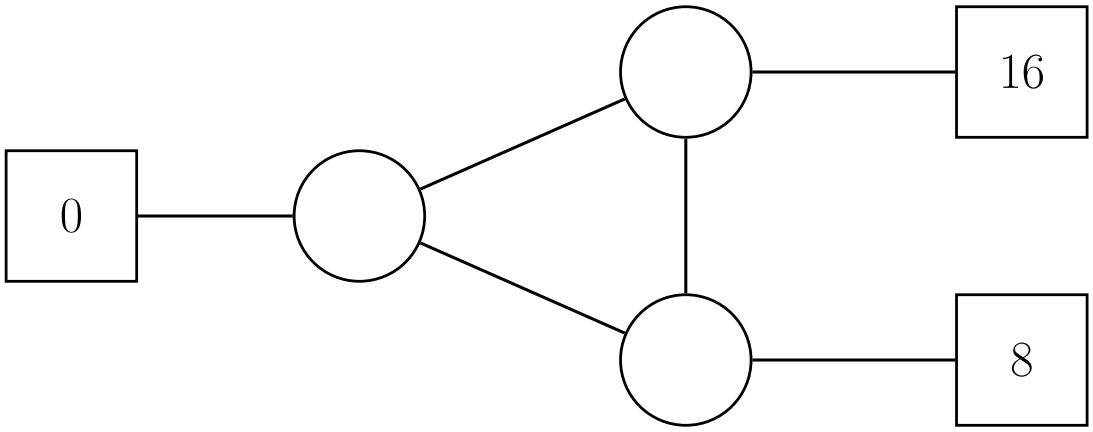
Problem 4: Fill the circles. Which circles can equal the largest square number on their own board? Explain what makes the two boards different.



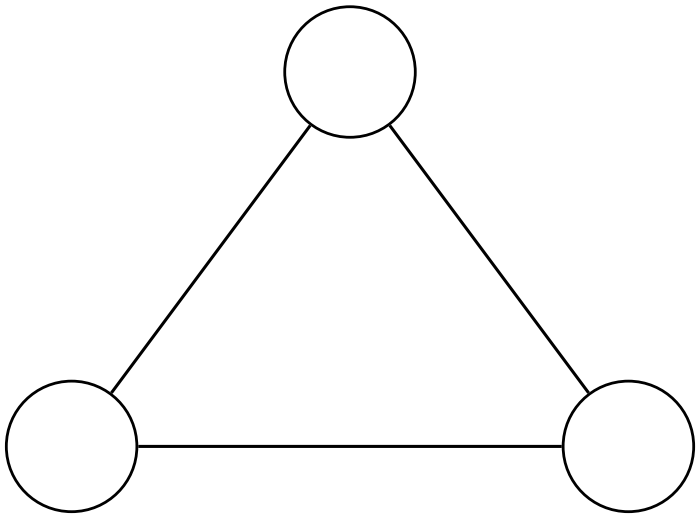


Problem 5: Can the same board, with the same square numbers, have two different fillings that both follow the rule? Explain your answer for every finite board where each shape can reach a square.





Problem 6: Use whole numbers from 0 to 4. Find every filling of the upper board and of the lower board. How many fillings does each board have? Explain why none are missing.





Problem 7: Which boards can you fill with whole numbers? Fill the other boards using the fraction cards.

